

Project Brief: Automated Talking Head Animation Pipeline

Project Overview

This is a standalone project. The objective is to build an automated pipeline that transforms static headshot images into realistic, natural-looking "talking" animations and programmatically sequences them with static "idle" states.

Since you will be working on a local machine without a dedicated GPU (e.g., a MacBook), you may leverage **free cloud GPU environments** (such as Google Colab or Kaggle Kernels) to run heavy deep learning inference models, while developing the orchestration, video processing, and sequencing scripts locally on your machine. You are free to use **any tools, libraries, APIs, or services** available to you — the approach is entirely up to you as long as the deliverables meet the requirements outlined below.

Objectives & Scope

You will be provided with **8 distinct headshot images** representing different individuals with varying facial structures, head poses, and lighting conditions. Your system must be robust enough to handle all 8 headshots without manual tuning per image.

The project is divided into three milestones:

Milestone 1: Natural Speaking Animation Generation

- **Goal:** Establish a repeatable process to generate high-quality, natural-looking talking animations from static headshots.
- **Requirements:**
 - Identify and utilize the best available tools or services (open-source models, commercial APIs, Hugging Face Spaces, existing Colab notebooks, or any other means) to generate talking head animations from static images. You may reuse existing tools and platforms — the goal is not to build from scratch, but to establish a **repeatable, documented process** that can be easily applied to new headshots in the future.
 - The generated talking animation does not need to match a specific spoken audio track perfectly, but it must look **naturally active** (realistic mouth movements, natural eye blinking, and subtle head/shoulder micro-movements).
 - The animation must be loopable or easily extendable to arbitrary lengths.

- **Robustness:** The pipeline must successfully generate high-quality animations for **all 8 provided headshots** without breaking or producing severe visual artifacts.
- **Repeatability:** The process must be clearly documented such that a non-technical team member could follow the steps to generate a new talking animation for any additional headshot in the future.

Milestone 2: Programmatic Video Sequencing Pipeline

- **Goal:** Build a local Python script that orchestrates the transition between "idle" (static) and "active" (talking) states based on timing parameters.
- **Requirements:**
 - Create a local Python CLI script that accepts three inputs:
 1. `static_headshot` (Path to the original high-resolution image)
 2. `speaking_animation` (Path to the animated video generated in Milestone 1)
 3. `S` (Speaking duration in seconds)
 - The script must programmatically generate a single, seamless output video in any popular animated format (e.g., MP4, GIF, WebM) with the following timeline:
 - **0.0s to 3.0s:** Static/Idle face (subtle breathing or blinking is a plus, but a clean static frame is the baseline).
 - **3.0s to (3.0 + S)s:** Active speaking animation (smoothly looping or stretching the Milestone 1 animation to fit exactly `S` seconds).
 - **(3.0 + S)s to (6.0 + S)s:** Return to static/Idle face.
 - **Transitions:** The transitions between the static state and the speaking state must be as smooth as possible (avoiding harsh jumps or sudden shifts in head position).

Milestone 3: Synchronized Audio + Animation Playback

- **Goal:** Build a playback system that simultaneously plays TTS-generated audio and displays the animated headshot, with automatic state transitions.
- **Requirements:**
 - Given a text paragraph as input, use a TTS service (e.g., OpenAI TTS, ElevenLabs, edge-tts, or any alternative) to generate a spoken audio file.
 - Build a playback system (e.g., a Python script with a GUI window, or a local HTML/JS demo page) that:
 1. Plays the TTS audio.
 2. Simultaneously displays the animated (talking) headshot in sync with the

audio playback.

3. When the audio playback ends, automatically returns the headshot to its static/idle state.

- Audio and animation must remain as **separate assets** — they are played in parallel, not baked/muxed into a single video file.
 - The transition from talking to idle when audio stops must be smooth and immediate.
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Technical Guidelines & Recommended Stack

You are encouraged to research and select the best tools for the job. The suggestions below are starting points, not constraints — feel free to explore alternatives, paid APIs with free tiers, or entirely different approaches.

1. Generative AI & Animation Models (Milestone 1)

Some open-source options that have shown strong results in this space include:

- **LivePortrait:** Highly efficient, fast inference, and produces extremely realistic facial expressions and structural stitching. It can be driven by a template video (which you can loop).
- **SadTalker:** A robust audio-driven 3D motion coefficient model. You can drive it using a generic "dummy" audio track or a silent track to generate natural-looking talking motions.
- **Wav2Lip:** Good for lip-syncing, though it may require more post-processing to keep the rest of the face looking natural.
- **Other options:** Any commercial API, Hugging Face Space, or emerging tool that achieves the desired result is acceptable.

2. Local Video Processing (Milestone 2)

For stitching, timing control, and video manipulation on your local MacBook Pro, common tools include:

- **OpenCV (python-opencv):** For frame-by-frame manipulation, resizing, and transition blending.
- **MoviePy:** A high-level Python library for video editing, useful for cutting, concatenating, and adjusting video durations.
- **FFmpeg:** The underlying CLI tool for efficient video encoding, codec management, and container wrapping.

- Again, you may use any alternative tools or libraries you prefer.

Project Timeline & Deliverables

All milestones will be evaluated by latest **July 2, 2026** (full-time start date).

Focus	Key Deliverables	Pay
Milestone 1: Natural Speaking Animation	Repeatable, documented process for generating talking animations. High-quality talking animations generated for all 8 headshots . Optimization of output to minimize visual artifacts. Clear step-by-step documentation for future use.	\$5,000
Milestone 2: Programmatic Sequencing Script	Python script implementing the [3s Static] → [S-sec Talk] → [3s Static] timeline. Smooth transition handling between idle and speaking states. Comprehensive README and dependency management.	\$4,000
Milestone 3: Synchronized Audio + Animation Playback	Playback system that plays TTS audio and animated headshot simultaneously. Headshot returns to static state when audio ends. Audio and animation remain as separate assets (not muxed into one file).	\$2,500

Note: Final payment for each milestone is discretionary and will be determined based on the degree of completion and the quality of outputs delivered.

Evaluation Criteria

Your project will be evaluated based on the following criteria:

1. **Visual Quality:** How natural does the speaking animation look? Are there noticeable warping artifacts, distortions, or sudden jumps during transitions?

2. **Robustness:** Does the pipeline work seamlessly across all 8 headshots, or does it fail on specific head poses/genders/lighting?
3. **Code Quality & Engineering:** Is the local Python script clean, modular, well-documented, and easy to run? Are dependencies properly managed (e.g., `requirements.txt` or `environment.yml`)?
4. **Repeatability & Documentation:** Can the process be easily followed by someone else to generate animations for new headshots without your involvement?